INTERNSHIP REPORT

ON "Summer Internship"

AT

Company: Techible under i3C in IIT Jammu

AN INDUSTRY INTERNSHIP REPORT SUBMITTED

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

SUBMITTED BY:

Harsh Vardhan Singh

COER UNIVERSITY

CANDIDATES' DECLARATION

I, Harsh Vardhan Singh, hereby declare that the work which is being presented in the Industry Internship Report entitled, "Summer Internship" in partial fulfillment of requirement for the award of degree of B.Tech, and submitted to COER UNIVERSITY, is an authentic record of my own work carried by me at Techible under i3c in IIT Jammu under the supervision and mentorship of Mr. Paramveer Nandal and Faculty Majid Ahmed respectively. The matter presented in this report has not been submitted in this or any other University / Institute for the award of Degree.
Signature of the Student: Harsh Vardhan Singh
Dated: 20/07/2026

INTERNSHIP CERTIFICATE

[Not Received Yet]

OFFER LETTER

SUMMER SCHOOL

INSTITUTE INCUBATION & INNOVATION COUNCIL I3C - IIT JAMMU

भारतीय प्रौद्योगिकी संस्थान जम्मू
INDIAN INSTITUTE OF TECHNOLOGY JAMMU

Techible

Offer Letter

To

Harsh Vardhan Singh

Re: Confirmation of Enrollment in Summer Internship Program

We are delighted to inform you that you have been successfully enrolled in the

Program: **IoT & Drones Building Program - 2M**

Offered at: **Indian Institute of Technology Jammu** in collaboration with **Techible** & **I3C-IIT Jammu**.

We congratulate you on your acceptance and are excited to have you as a part of our program. This internship will be an excellent opportunity for you to enhance your expertise in the enrolled domain and is expected to help you in next phases of your career.

The internship program will commence from **June 10, 2026** following the completion of necessary administrative procedures.

As a participant, you will have access to expert mentors and a collaborative environment. You will work on various projects (at times in a team), engage in interactive sessions, and receive guidance from professionals and academic experts.

Congratulations once again, and we wish you a successful and enriching internship experience.

Regards,
Team Summer School

Phone +91-9186184541

Email summerschool@iitjammu.ac.in

Website summerschool.iitjammu.ac.in

ACKNOWLEDGEMENTS

Industry Internship is an important aspect in the field of engineering, where contribution is made by many persons and organizations. The present shape of this work has come forth after contribution from different spheres.

- I Give acknowledgement to Paramveer Nandal Supervisor.
- I Give acknowledgement to Majid Ahmed Supervisor.
- I Give acknowledgement to Head, Training and Placement and others.
- I Give acknowledgement to H.O.D, Mrs. Surbhi Sharma.
- I Give acknowledgement to Director, Mr. Ankur Gupta.
- I would also like to thank my parents, friends etc. who helped me in my Industry Internship.
- I express my sincere gratitude to Techible under i3c in IIT Jammu for giving me the opportunity to work on Industry Internship during my degree.
- I express my sincere gratitude to COER University for giving me the opportunity to work on Industry Internship.
- At the end thanks to the Almighty for everything.

CONTENTS

ABBREVIATIONS USED

- **ANN** - Artificial Neural Network
- **BLE** - Bluetooth Low Energy
- **HTTP** - Hypertext Transfer Protocol
- **I2C** - Inter-Integrated Circuit
- **IoT** - Internet of Things
- **MQTT** - Message Queuing Telemetry Transport
- **SPI** - Serial Peripheral Interface
- **UART** - Universal Asynchronous Receiver Transmitter

CHAPTER 1: INTRODUCTION

1.1 About Techible IIT Jammu Summer School Internship

The IoT Summer School Internship 2026, organized by Techible under i3c in IIT Jammu, was designed to provide undergraduate engineering students with practical exposure to the rapidly growing field of the Internet of Things (IoT) and embedded system development. The internship was structured as an intensive 42-day learning program from 10 June to 20 July combining theoretical lectures, laboratory experiments, assignments, and project-based learning. Rather than focusing only on programming concepts, the internship emphasized the integration of hardware and software to develop intelligent systems capable of sensing, processing, and communicating information over computer networks.

With the increasing adoption of automation, Industry 4.0, smart cities, connected healthcare, and intelligent transportation systems, IoT has emerged as one of the most significant technological domains of the modern era. The internship introduced students to the fundamental principles governing IoT devices, including microcontrollers, communication protocols, embedded programming, cloud computing, wireless networking, and sensor interfacing. Each laboratory session was carefully designed to bridge the gap between classroom theory and practical implementation by allowing students to build and test real-world embedded applications.

1.2 Objectives of the Internship

The primary objective of the internship was to provide students with practical knowledge of IoT technologies while strengthening their understanding of embedded systems and modern communication techniques. The internship aimed to develop problem-solving skills by encouraging students to implement hardware-based solutions rather than merely studying theoretical concepts.

The major objectives of the internship are summarized below:

- To understand the fundamentals of Internet of Things technology and its industrial applications.
- To develop programming skills using the Arduino Integrated Development Environment (IDE) and Node.js for backend scripting.
- To study the architecture and working principles of embedded systems including the ESP32 and Raspberry Pi.
- To gain hands-on experience with embedded development boards and operating systems like Ubuntu.
- To understand various communication protocols including UART, SPI, I2C, BLE, MQTT, and HTTP.
- To interface different sensors and actuators, specifically utilizing I2C bus communications, and resolve hardware grounding issues.
- To develop a unified web dashboard hosted locally on port 3000 to monitor and control IoT devices.
- To understand software version control using Git and GitHub for hosting assignments and project code.
- To improve technical documentation and report writing skills.

1.3 Internet of Things (IoT)

The Internet of Things (IoT) refers to a network of interconnected physical devices capable of collecting, processing, and exchanging information through communication networks without requiring continuous human intervention. These devices contain embedded processors, sensors, communication modules, and software that enable them to interact intelligently with their surroundings.

Unlike traditional computer systems, IoT devices continuously monitor environmental conditions, collect

real-time data, process the information locally or remotely, and perform predefined actions based on programmed logic. Such intelligent behavior has transformed numerous industries by enabling automation, predictive maintenance, resource optimization, and remote monitoring.
A typical IoT system consists of several functional components. Sensors collect environmental data. The collected data is processed by a microcontroller such as the ESP32 or a single-board computer like a Raspberry Pi, which performs logical operations and prepares the information for transmission. Communication modules establish connectivity through Wi-Fi, Bluetooth, or local bus networks. Finally, local or cloud servers store and analyze the transmitted data while providing remote visualization through web applications or mobile devices.

1.4 Applications of IoT

The Internet of Things has become one of the most influential technologies across almost every engineering discipline. Its ability to connect physical devices with digital networks has revolutionized traditional methods of monitoring, automation, and decision-making.

- **Smart Homes:** Smart home automation represents one of the most recognizable applications of IoT technology. Intelligent lighting systems, smart locks, and surveillance cameras communicate through wireless networks to improve convenience, security, and energy efficiency.
- **Smart Healthcare:** Healthcare has greatly benefited from IoT-based monitoring devices capable of continuously recording physiological parameters. Wearable devices transmit patient information directly to hospitals, allowing early diagnosis and reducing emergency response time.
- **Industrial Automation:** Industrial IoT enables factories to monitor machinery continuously using sensors that measure vibration, temperature, pressure, and electrical current. Predictive maintenance algorithms analyze collected data to identify abnormal operating conditions before equipment failure occurs.
- **Smart Cities:** Smart cities integrate multiple IoT applications including intelligent street lighting, traffic management, waste collection, parking management, environmental monitoring, and public safety systems.

1.5 Internship Schedule

The internship was conducted over a period of 42 days, focusing on progressively advanced concepts in embedded systems, web development, and IoT integration.

- **Phase 1: Hardware and Environment Setup.** Introduction to embedded programming, familiarization with microcontrollers, OS installations (Ubuntu), and Git/GitHub setup.
- **Phase 2: Embedded Communication Protocols.** Focus on communication between electronic devices. Studying UART, SPI, and I2C protocols, and completing assignments on sensor interfacing.
- **Phase 3: Web Development and IoT Connectivity.** Starting backend development using Node.js to set up a localhost server environment. Developing frontend UI and wiring it to backend API endpoints.
- **Phase 4: Final Project Integration.** Troubleshooting hardware issues (such as camera modules and I2C bus configurations), finalizing the local web dashboard, and preparing the technical internship report.

CHAPTER 2: THEORETICAL BACKGROUND OF IoT AND EMBEDDED SYSTEMS

2.1 IoT Architecture

The Internet of Things (IoT) architecture defines the structural framework through which interconnected devices collect data, communicate with one another, process information, and deliver useful services to users. A well-designed IoT architecture ensures reliable communication between hardware devices, local servers, and end-user applications while maintaining scalability, security, and efficient resource utilization. Modern IoT systems consist of multiple layers, each responsible for a specific function in the overall data communication process.

Perception Layer: The Perception Layer is the physical layer of an IoT system. It consists of sensors, actuators, RFID tags, cameras, GPS modules, temperature sensors, humidity sensors, gas sensors, ultrasonic sensors, and various electronic devices responsible for interacting with the surrounding environment. Sensors continuously monitor physical parameters and convert them into electrical signals that can be processed by a microcontroller.

Network Layer: The Network Layer is responsible for transmitting data collected by sensors to remote servers or cloud platforms. It acts as the communication bridge between embedded hardware and software applications. Common communication technologies include Wi-Fi, Bluetooth Low Energy (BLE), ZigBee, LoRaWAN, GSM/4G/5G, Ethernet, MQTT, HTTP, and CoAP.

Processing Layer: The Processing Layer is responsible for storing, analyzing, and processing the information received from IoT devices. Data collected from multiple sensors may be stored either locally within edge computing devices or remotely on cloud platforms.

Application Layer: The Application Layer is the user interface of the IoT system. It presents processed information in a meaningful format through websites, mobile applications, dashboards, or desktop software.

2.2 Embedded Systems

An Embedded System is a dedicated computer system designed to perform one or more specific tasks within a larger mechanical or electrical system. Unlike general-purpose computers such as desktops and laptops, embedded systems are optimized for particular applications and are programmed to execute predefined operations efficiently with minimal hardware resources.

A typical embedded system consists of the following major components:

- **Microcontroller/Microprocessor:** The central brain that executes programmed instructions stored in memory and controls all connected peripherals.
- **Memory:** Memory stores both the program instructions and temporary runtime data.
- **Input Devices:** Input devices collect information from the environment (e.g., temperature sensors, cameras, accelerometers).
- **Output Devices:** Output devices receive control signals and perform useful actions (e.g., LEDs, Displays, Motors).

2.3 Microcontroller vs Microprocessor

A microprocessor consists primarily of a Central Processing Unit (CPU) that requires external memory, input/output controllers, timers, communication peripherals, and storage devices to function as a complete computer system (e.g., Raspberry Pi utilizes a microprocessor architecture). A microcontroller, on the other hand, integrates the CPU, RAM, Flash Memory, timers, GPIO pins, communication interfaces, and interrupt controllers onto a single integrated circuit (e.g., ESP32).

2.4 Communication Protocols

Communication protocols define standardized rules governing data exchange between electronic devices. Without standardized communication methods, different hardware components would be unable to exchange information reliably.

- **UART (Universal Asynchronous Receiver Transmitter):** One of the simplest methods of serial communication, using two primary wires: TX (Transmit) and RX (Receive).
- **SPI (Serial Peripheral Interface):** A high-speed synchronous communication protocol used when rapid data transfer is required, using MOSI, MISO, SCK, and CS.
- **I2C (Inter-Integrated Circuit):** Designed to reduce wiring complexity, using only two wires: SDA (Serial Data) and SCL (Serial Clock). Multiple slave devices can communicate with a single master using unique addresses, which was critical for sensor interfacing in this internship.
- **MQTT & HTTP:** Lightweight messaging and web standard protocols utilized to route data between the hardware and the web dashboard.

2.5 ESP32 Architecture

The ESP32 is a powerful, low-cost, and energy-efficient microcontroller developed by Espressif Systems for Internet of Things applications. It combines high processing capability with integrated Wi-Fi and Bluetooth connectivity, making it one of the most widely used development platforms for embedded and IoT projects. It features dual-core processing, PWM Channels, SPI, I2C, and UART interfaces, alongside Cryptographic Hardware Accelerators.

CHAPTER 3: DEVELOPMENT TOOLS, COMMUNICATION PROTOCOLS AND SENSOR INTERFACING

3.1 Arduino IDE and Hardware Tooling

The Arduino Integrated Development Environment (Arduino IDE) is an open-source software platform used for developing programs for various microcontrollers. In conjunction with standard microcontroller IDEs, advanced tooling was utilized for custom printed circuit board (PCB) design and schematic mapping using platforms like EasyEDA Pro. Furthermore, embedded device environments such as Ubuntu OS were leveraged on microcomputing hardware to host complex backend services and interface with advanced peripherals like camera modules.

3.2 Communication Protocols and Practical Implementation

Modern embedded devices exchange information continuously. During the 42-day internship, a major focus was placed on practical communication, specifically utilizing the I2C bus. Scripts written in Node.js and Python were developed to acquire raw JSON data from connected hardware peripherals over the I2C protocol. This enabled compact embedded systems to efficiently support multiple sensing devices while minimizing physical wiring.

3.3 Sensor Interfacing

A sensor detects physical parameters and converts them into electrical signals. These signals are read via GPIO pins. During the internship, various peripheral sensors and camera modules were interfaced with the core processing unit. The data, once acquired, was processed and transmitted through local network routing (HTTP APIs) rather than relying exclusively on cloud infrastructure, ensuring low latency and high security for the local dashboard application.

CHAPTER 4: PROJECT WORK & WEBSITE DASHBOARD

4.1 Project Introduction

Problem Statement: Managing and visualizing data from disparate IoT sensors locally requires a unified, responsive interface that operates with minimal latency.
Objectives: To design, develop, and deploy a local web application that seamlessly reads and displays data from hardware devices in real time, serving as a comprehensive IoT dashboard.
Scope: Hardware data acquisition, local backend server routing, and frontend UI design.

4.2 System Design & Website Integration

Overall Architecture: The project follows a robust 3-tier architecture:

1. **Hardware Layer:** Embedded devices and sensors connected via I2C and UART.
2. **Backend API:** A Node.js application running on the host machine processing incoming hardware telemetry.
3. **Frontend Web Dashboard:** A responsive web interface served to the browser.

Workflow: Physical devices collect metrics -> The metrics are transmitted to the central processing unit -> The Node.js backend serves these metrics through an API -> The browser dashboard fetches and dynamically visualizes the data.

4.3 Technology Stack

- **Backend:** Node.js, Express.js
- **Frontend:** HTML, CSS, JavaScript (Fetch API for real-time updates)
- **Embedded Hardware Scripts:** Python, C++
- **Operating System:** Ubuntu
- **Version Control:** Git & GitHub

4.4 System Implementation & GitHub Deployment

Module 1: Hardware Interfacing & Data Collection

Purpose: Gather peripheral data accurately.
Implementation: Python and Node scripts utilizing the I2C bus to poll sensors at regular intervals.
Output: Raw JSON data output.

Module 2: Backend API (Node.js)

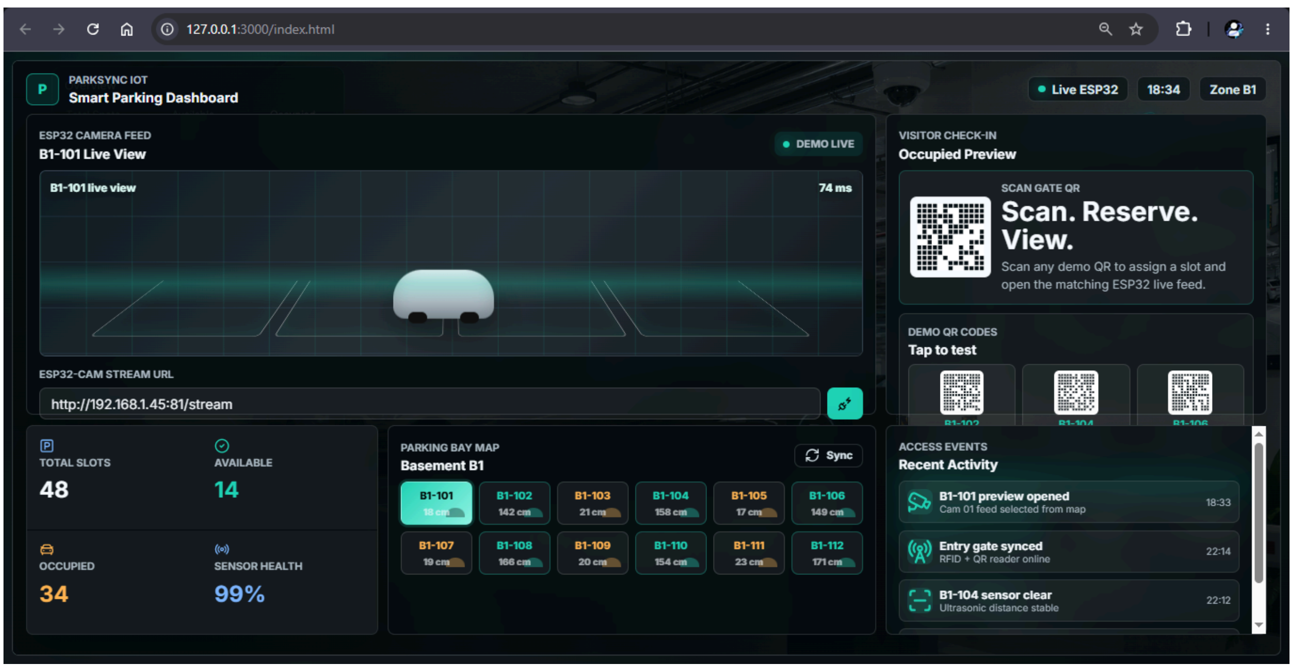
Purpose: Act as a local bridge between the hardware network and the frontend user interface.
Implementation: An Express server deployed and running on localhost port 3000.
Output: Successful REST API endpoints that serve both the static web files and the dynamic sensor data

feeds.

Module 3: Frontend Web Dashboard

Purpose: Visualize the telemetry for the end-user in an intuitive manner.
Implementation: A highly responsive web layout that fetches real-time data from the Node API endpoints without requiring page reloads.
Output: A functional dynamic UI deployed locally at localhost:3000/index.html.
All source code, assignments, and iterative project milestones were successfully committed and maintained in a professional GitHub repository specifically created for this internship program.

Project Screenshot:-



Img 1

CHAPTER 5: WEEKLY WORK LOG & PRACTICAL OUTPUT

5.1 Week-wise Breakdown (6 weeks)

Phase / Week	Summary of Activities
Week 1-2	Orientation under i3c in IIT Jammu. Environment setup, OS configuration (Ubuntu), GitHub repository initialization. Began practical assignments on fundamental hardware familiarization and studied power delivery layouts.
Week 3-4	Completed extensive assignments on sensor interfacing via the I2C bus. Commenced backend development using Node.js. Successfully set up the Express server environment to listen on port 3000.
Week 5-6	Frontend UI development of the IoT dashboard. Wired the frontend to backend API endpoints for real-time rendering. Handled intensive troubleshooting and integration testing. Prepared final documentation and pushed all code to GitHub.

5.2 Challenges Faced & Resolutions

- **Technical Challenges:** Encountered video stream errors (grey screen) when configuring a specific camera module on the Ubuntu operating system. Additionally, diagnosing the exact device addresses on the I2C bus presented initial connectivity hurdles.
- **Grounding & Circuitry:** Navigated power grounding distinctions while mapping the hardware network. It was vital to separate analog and power grounding to ensure clean signal integrity for the sensitive sensors.
- **Problem Solving:** Relied heavily on deep technical datasheets, reviewing Ubuntu kernel logs, and utilizing I2C bus detection tools to isolate and resolve connectivity issues iteratively.

CHAPTER 6: CONCLUSIONS

6.1 Learning Outcomes and Future Scope

Conclusion: The 42-day IoT Summer School Internship under Techible at IIT Jammu was highly successful. It culminated in a robust, locally hosted web application that interfaces directly with custom IoT hardware assignments. Through hands-on experimentation, rigorous debugging, and full-stack project development, theoretical concepts were transformed into practical engineering solutions.

Learning Outcomes: Practical mastery over embedded programming, sensor interfacing via I2C, Node.js backend API creation, web dashboard UI design, and professional source code management via GitHub. The experience provided a comprehensive understanding of bridging low-level hardware engineering with high-level software development.

Future Scope: Future iterations of the project could involve transitioning the localhost server to a cloud infrastructure platform (such as AWS or GCP) for remote geographic monitoring. Additionally, integrating artificial intelligence and predictive analytics models to process the gathered sensor telemetry could elevate the system from a monitoring tool to an intelligent, automated decision-making platform.

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5. Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M. and Ayyash, M., 2015, "Internet of Things: A Survey on Enabling Technologies, Protocols and Applications", IEEE Communications Surveys & Tutorials, Volume 17, Issue 4, pp. 2347-2376.
6. Banks, A. and Gupta, R., 2014, MQTT Version 3.1.1, OASIS Standard.

APPENDICES

Appendix A: Internship Offer Letter
Appendix B: Internship Completion Certificate
Appendix C: Project Screenshots [img1]
Appendix D: Source Code Snippets and Repository Reference
(https://github.com/HARSHVARDHANSINGH1903/iot-summer-school-internship-2026)